

Summary of Penetration Testing Report: Attack from End User's Desktop

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The penetration testing report, "Attack from End User's Desktop," conducted by Jordan Flaum, Rafey Youniacutt, and Mung Mang, focuses on identifying and addressing security vulnerabilities within Costco's network. The report outlines a simulated attack originating from an end user's desktop to evaluate the company's defenses and response mechanisms.

Conclusion:

To enhance security and prevent similar attacks in the future, Costco should implement several critical upgrades:

1. **Data Input Validation:** Adopting rigorous validation practices can defend against SQL injection attempts.
2. **Anti-Phishing Training:** Ongoing and focused training for employees will improve their ability to recognize and resist social engineering tactics.
3. **Advanced Email Security:** Deploying sophisticated email security solutions and multi-factor authentication will strengthen network defenses.
4. **Regular Audits and Patch Management:** Frequent security audits and prompt application of patches are essential to maintain a strong security posture.
5. **Cybersecurity Awareness:** Cultivating a culture of cybersecurity awareness is crucial for mitigating risks from evolving cyber threats.

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Executive Summary

This executive summary explains the core findings and strategic insights derived from a penetration testing project conducted on the cybersecurity infrastructure of Costco Wholesale Warehouse. The objective of this simulated attack was to assess vulnerabilities and exploit lax security practices within the organization, specifically targeting negligent supervisors and managers who left their emails open and USB ports enabled on accessible desktops. The penetration testing project focused on compromising an end user's device to illicitly obtain credit card information and unauthorized access to sensitive emails, leveraging multiple attack vectors and strategic pivots.

Introduction

In an era where cybersecurity threats are escalating, the heightened demand to fortify and secure the digital fortresses of organizations has become paramount. This penetration testing project at a Costco Wholesale Warehouse emerges as a crucial endeavor aimed at evaluating and fortifying the security posture of this retail giant. Recognizing that the end user's laptop often serves as the gateway to an organization's livelihood and functionality, our focus on this entry point underscores the necessity for a comprehensive analysis. As technology advances, so do the tactics of malicious actors seeking to exploit vulnerabilities. By delving into the potential vulnerabilities originating from an end user's device, we seek not only to identify weaknesses but also to provide strategic recommendations for bolstering Costco's defenses against evolving cyber threats. This penetration testing project aligns with the proactive ethos necessary for safeguarding sensitive data, ensuring the resilience of critical operations, and fortifying the trust of both employees and members in the face of an ever-evolving digital landscape.

Background

Costco Wholesale Warehouses, positioned at the pinnacle of the retail industry, operate within a dynamic landscape marked by impressive financial figures and a distinctive business model. With a staggering revenue of \$242.29 billion in the twelve months ending September 3, 2023, boasting a remarkable 6.76% year-over-year growth, Costco has established itself as a retail powerhouse (Stock Analysis, 2023). The foundation of its success lies in the operation of membership warehouses and e-commerce platforms, guided by the principle of offering members compelling value through low prices on a select range of nationally branded and private-label products. Leveraging high sales volumes, rapid inventory turnover, and operational

efficiencies, Costco maintains profitability at notably lower gross margins than its retail counterparts.

Critical to its operations are streamlined merchandise receiving and distribution processes, primarily managed through depots, and a network of facilities supporting processing, packaging, manufacturing, and the production of private-label items. However, the vulnerability of these operations to disruptions, ranging from natural disasters to pandemics, poses a significant risk to the orderly flow of merchandise to warehouses. Additionally, Costco's e-commerce operations, integral to its business strategy, heavily rely on third-party and in-house logistics providers, creating a susceptibility to delays and disruptions. (Costco Wholesale, 2022)

The rationale for choosing Costco as the subject of this penetration testing project stems from an insider's perspective. As a current Costco employee, I have observed instances of negligence among staff regarding security measures, potentially exposing the organization to breaches and data theft. This project aims to address and mitigate these vulnerabilities, ensuring the resilience of Costco's digital fortifications and the preservation of trust among its valued members.

Attack Objective

The primary goal of this penetration testing endeavor is to simulate and analyze the potential risks associated with a USB-based attack, specifically targeting end-user desktops within the Costco network. The objective is to gain unauthorized access to sensitive data, uncovering vulnerabilities that, if exploited by malicious actors, could compromise the organization's security posture. Costco, as a retail giant, was chosen strategically for this penetration test due to its expansive operations and the significant volume of sensitive data it manages. Retail businesses like Costco handle vast amounts of customer information, financial

records, and proprietary data. By selecting Costco, the test aims to evaluate the resilience of a business operating on such a scale, where security lapses could have severe consequences.

The focus is on acquiring access to customer and employee data stored within the Costco network. This includes personally identifiable information, financial records, and other confidential data that, if compromised, could lead to identity theft, financial fraud, or other malicious activities. The significance of the targeted data lies in its potential for exploitation by malicious actors. Customer data can be monetized on the dark web, leading to identity theft, fraudulent transactions, and compromise of individuals' privacy. Employee data can be leveraged for spear-phishing attacks, and financial records may be exploited for financial gain. By aiming for such critical data, this penetration test seeks to highlight the potential real-world repercussions of a successful attack on Costco's systems. (F-Secure, 2023)

Understanding the significance of the data underscores the importance of robust cybersecurity measures in place to protect against unauthorized access and potential breaches, safeguarding the trust of both customers and employees. This simulation serves as a proactive measure to identify and rectify vulnerabilities, ensuring Costco's ongoing commitment to data security and maintaining the integrity of its operations.

Expected Attack Response

In the simulated penetration test targeting Costco's end-user desktops through USB exploitation, the expected attack response hinges on the success or failure of the devised strategies. If the security measures in place prove robust and the end users demonstrate a high level of awareness, the attack is likely to face resistance, resulting in failure. In such a scenario, the USB device may trigger security alerts, and the end user might recognize the phishing attempt or the malicious payload, thereby thwarting the unauthorized access attempt.

Conversely, if the attack is successful, it signifies potential vulnerabilities within Costco's security infrastructure. Successful infiltration may be indicative of lapses in security awareness among end users or shortcomings in endpoint protection. The USB-based compromise may go unnoticed, allowing the attacker to gain a foothold on the end user's desktop. This success paves the way for further exploration and potential lateral movement within Costco's network, emphasizing the critical need for robust security measures and ongoing user education to mitigate such threats effectively. The expected attack response, whether success or failure, serves as a litmus test for Costco's current security resilience and highlights areas that require reinforcement to safeguard against real-world cyber threats.

Expected Barriers

Attempting to gain access to Costco's internal systems via a USB device encounters several expected barriers that are crucial components of the organization's security measures. One notable barrier is the potential use of two-factor authentication (2FA), specifically PingID, which adds an additional layer of protection beyond a simple username and password. While 2FA enhances security, determined attackers may employ various tactics to navigate around it. (Ping Identity, 2020)

The primary challenge arises in bypassing or evading the PingID 2FA. This may involve exploiting vulnerabilities in the authentication process, social engineering techniques to trick users into divulging authentication codes, or deploying advanced phishing attacks that target both login credentials and 2FA codes simultaneously. Attackers might also attempt to compromise the end user's device directly, seeking to intercept or manipulate the 2FA process before it reaches the authentication server.

Additionally, Costco's security infrastructure likely includes endpoint protection solutions, which can pose another significant barrier. Traditional antivirus programs and endpoint detection tools may recognize the USB device as a potential threat, triggering alerts or blocking malicious activities. The attacker must therefore employ evasion techniques, such as using sophisticated malware variants or manipulating the USB device's characteristics to avoid detection. (OSIbeyond, 2023)

Another notable factor, is employee awareness and vigilance form a critical barrier. If end users are well-informed about security best practices, they may recognize suspicious USB devices or phishing attempts, reducing the chances of a successful attack. Therefore, social engineering becomes a key strategy for attackers, attempting to exploit human vulnerabilities to override technical barriers. (Tripwire, 2023)

In summary, the expected barriers involve overcoming 2FA through various means, evading endpoint protection solutions, and navigating the challenges posed by informed and vigilant end users. The success or failure of the penetration test depends on the effectiveness of the attacker in overcoming these hurdles and exposing potential weaknesses in Costco's security defenses.

How the Barriers Are Defeated

To defeat the barriers encountered during the penetration test aimed at gaining access to Costco's internal systems via a USB device on an enabled port, we must employ a combination of technical exploits, social engineering tactics, and evasion techniques.

My plan of attack involves employing a USB device drop attack to exploit vulnerabilities in Costco's security and capture the handshake from the 2FA PingID and user credentials using a

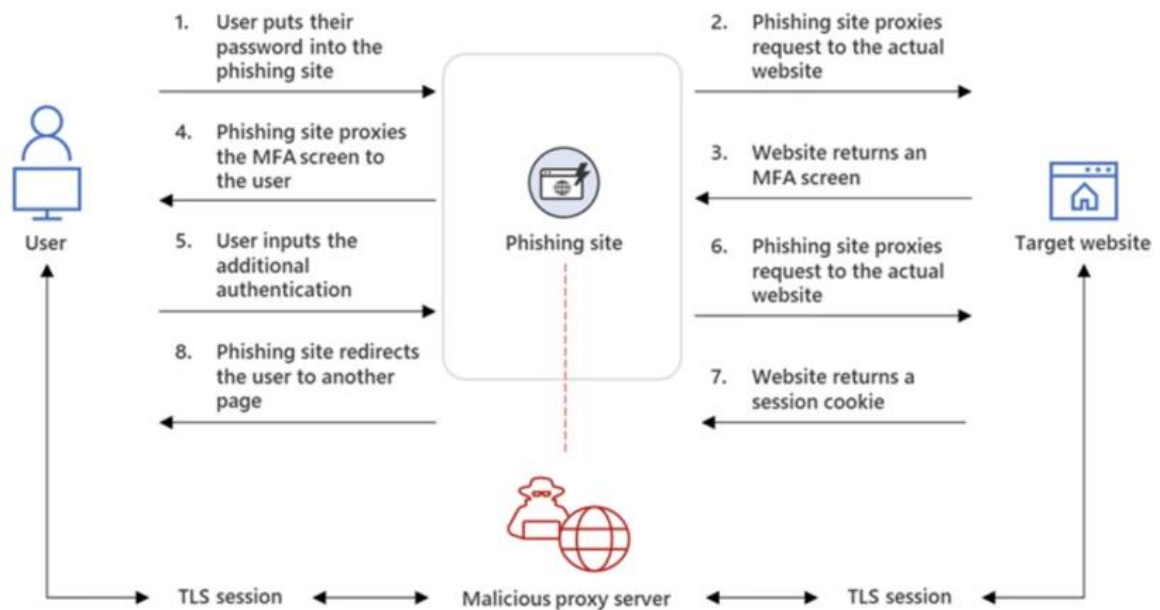
session cookie attack that will enable a bypass of the 2FA. The attack is carefully orchestrated to take advantage of physical access to a front-end desktop accessible to passing members. The attack leverages the Rubber Ducky, a powerful tool capable of emulating keyboard inputs, allowing for rapid execution of predefined scripts.

The USB Rubber Ducky is a versatile hacking tool developed by Hak5, designed to exploit vulnerabilities and automate various tasks through keystroke injection attacks (Hak5, 2023). Its primary purpose is to mimic a USB keyboard, allowing it to inject predefined keystrokes into a target system. While the tool is widely used for legitimate security testing and educational purposes, it's essential to note that it can be misused for malicious activities. The Rubber Ducky can emulate a keyboard, enabling the injection of pre-programmed keystrokes into a target system. This feature allows for the automation of complex tasks or the execution of specific commands. Predefined payloads can be loaded onto the Rubber Ducky to execute specific actions on the target system. This can include running malicious commands, downloading, and executing files, or even setting up persistent backdoor access. The USB Rubber Ducky, being a hardware-based attack tool, can potentially bypass certain endpoint detection mechanisms. The Rubber Ducky operates at a low level, emulating a keyboard at the hardware level. Traditional endpoint security solutions may focus more on software-based threats, making it challenging to detect Rubber Ducky's activities. (Geeks for Geeks, 2023)

To conduct this attack, we will be loading the Rubber Ducky USB with Evilginx 2. Evilginx is a tool used for phishing attacks (Hackmag, 2023). Instead of imitating sign-in pages, Evilginx acts as a go-between for the real website and the targeted user. The user interacts with the actual website, but Evilginx secretly captures all the information exchanged between them. By functioning as a middleman or a man-in-the-middle attack, Evilginx not only gathers

usernames and passwords but also captures authentication tokens transmitted as cookies.

Evilginx uses what are called islets or small configuration files used for targeting specific websites. (EvilGinx, 2023)



(Diagram Bypassing 2FA Authentication with Evilginx2, 2022)

Our goal is to plug in the USB device to one of the two desktops that are accessible as you walk by the member service station right next to the bathrooms. This is a very high-traffic area since there is only one member restroom area. These desktops are used by employees, supervisors, and managers for looking up items and accessing emails, schedules, etc. We need to walk by insert our device into one of the accessible USB ports and continue walking to the bathroom. After plugging in our device, it will run its script programs to automatically run Evilginx. After leaving the bathrooms all we need to do is simply act like we are shopping and wait for a user to use the desktop and sign into their email or employee account. When a supervisor or manager accesses their email or employee account Evilginx will capture the session

cookie and we will be alerted on our device, in this case, our iPhone. We will be able to simply take the captured cookie and paste it into the Google cookie editor (Google Chrome, 2023). This extension will allow us to take the cookie and use its token to sign into our 2FA and we will have direct access to the network along with that supervisor or manager's account credentials. From this point, we can pivot our attack.

Once the attackers have successfully bypassed the login and 2FA process, the next step is to pivot towards the collection of information. We would aim to harvest as much information as possible and permission to escalate to acquire more sensitive data. Primary targets may include information such as logins, personally identifiable information, financial records, and other confidential data with the potential for financial gain. After collecting this data, we would look to sell it to buyers or brokers on the dark web. Additionally, we could leverage the compromised 2FA account depending on what account we can gain access to, to continue our attack by utilizing strategies like phishing emails and social engineering to escalate permissions and acquire even more sensitive information.

Accessing a supervisor or manager's account allows us to explore other network systems and accounts, mapping out the network infrastructure and identifying potential targets. We can search for shared drives, network shares, or privileged accounts with access to critical systems or databases. By leveraging the compromised account, we can gain access to these resources and expand our control over the network. This allows us to explore higher-level accounts or systems with more sensitive information, potentially gaining access to the network. A compromised account at Costco can be used for social engineering attacks, aiming to trick employees into revealing sensitive information. This can be done through phishing emails or messages, which appear to be from a trusted individual (Allen, 2023). The goal is to gain access to additional user

accounts, expand network control, and collect valuable information. The attackers can also trick employees into clicking on malicious links or downloading harmful files. This can help in infiltrating the network and gaining access to additional resources.

In addition to lateral movement, we can also exploit privilege escalation to gain higher-level access within the network. This involves identifying vulnerabilities, misconfigurations, or weak access controls that allow us to elevate our privileges. For instance, we can exploit software or operating system vulnerabilities to gain administrative access, manipulate user permissions, or exploit weak password policies (BeyondTrust, 2023). By escalating privileges, we can bypass security measures and access more critical systems, databases, or sensitive data. After gaining access to sensitive data, the next step is to exfiltrate it without detection. This can be achieved through techniques like encryption, steganography, or using covert channels to transmit the data outside the network. Encrypting the data to prevent detection by network security tools, using steganography to hide the data within seemingly innocent files, and establishing covert communication channels to transmit the data discreetly can help prevent future exploitation and leave traceable evidence behind. This ensures the potential for future exploitation and avoids leaving traceable evidence behind.

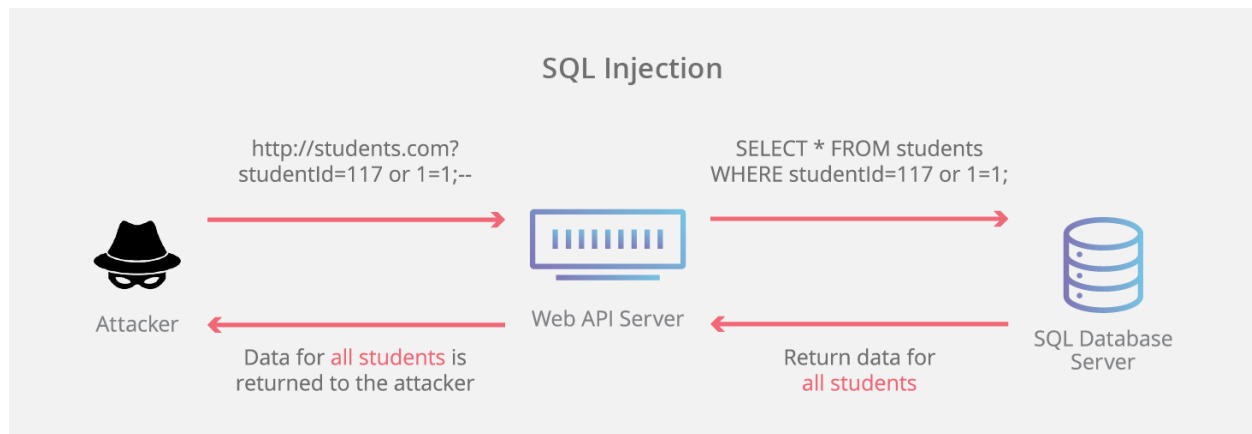
Attackers can use spear-phishing techniques to dramatically escalate permissions and gain access to additional systems within Costco's network. They can trick individuals into revealing sensitive information or clicking on malicious links or attachments just like with traditional phishing tactics. The difference with spear phishing is that regular phishing casts a wide net with a large number of targets while spear phishing is a customized attack designed for a specific person in mind, typically someone much higher up in the company that would result in much greater yields of permissions and information. Targets could include the Network Access

Control (NAC) server, Video Surveillance server, Web server, Email server, Small Data Center or Server Farm, and Wireless LAN Controller.

An attacker can impersonate a trusted IT administrator or security personnel, send emails requesting login credentials or software updates, and gain control over the NAC server. They can manipulate network access and potentially bypass security measures. The Video Surveillance server can be manipulated or disabled, allowing the attackers to move undetected. The Web server can be targeted by posing as a member of the IT department or a trusted service provider, allowing them to deface the website, inject malicious code, or exfiltrate sensitive data. The Email server can be targeted by posing as an important business contact or IT department member, allowing them to gain control over sensitive emails. The Wireless LAN Controller can be targeted by posing as a wireless network equipment vendor or security team member, allowing the attackers to intercept sensitive data transmitted over Wi-Fi or launch further attacks on connected devices.

The other effective type of attacking Costco network is SQLI. SQL injection attacks have undeniably become one of the most insidious and prevalent online threats in the last decade. These attacks represent a sophisticated strategy employed by malicious actors to compromise the security of servers, exploiting vulnerabilities in input fields commonly found in online forms. The crux of SQL injection lies in the injection of malicious scripts into these input fields, a method that aims to manipulate and deceive servers into providing unauthorized access to sensitive database information (Tripwire, 2023). The potential fallout from successful SQL injection attacks is extensive. Attackers can compromise critical elements of server infrastructure, including cookies, web forms, and HTTP posts. This not only jeopardizes the

confidentiality and integrity of data but also opens the door for manipulation of information stored in databases.



Through SQL injection (SQLi), the assailant exploits a vulnerable Application Programming Interface (API) as the entry point. The API serves as the software interface facilitating communication between a server and the requests it receives (Malwarebytes, 2023). SQL injection attacks targeting APIs are particularly insidious due to their potential to manipulate databases by injecting malicious SQL queries.

In the clandestine pursuit of exploiting Costco's digital fortress, our adversary deftly deploys sophisticated tools designed to navigate the intricacies of web architecture. These readily available tools are crafted with automation in mind, enabling the attacker to systematically scan the vast expanse of the Costco website with unprecedented efficiency. Their primary objective is to identify the elusive forms and Application Programming Interface (API) endpoints that serve as potential gateways into the system's underbelly. Once these crucial entry points are pinpointed, the tools execute a meticulously orchestrated sequence (OWASP Foundation, 2023). Armed with a repertoire of SQL queries, the attacker systematically probes the identified forms and API endpoints. This insidious dance is a delicate balancing act, aimed at exploiting potential

vulnerabilities in the website's code. The queries injected into the system are carefully crafted to manipulate the database in ways the website's developers did not foresee, opening doors to unauthorized access.

The probing process is akin to a digital cat-and-mouse game, anticipating and responding to the system's reactions to their injected queries. The goal is to elicit responses that go beyond the expected, revealing unintended behaviors or weaknesses in the database architecture. This methodical approach is designed to uncover exploitable flaws, the discovery of which would offer a nefarious gateway for the attacker to infiltrate Costco's database. The potential fallout from this method is grave. Gaining unauthorized access provides the attacker with a virtual key to Costco's trove of sensitive information. The compromised database, typically a repository for customer data, financial records, and other confidential details, becomes a treasure trove for exploitation (*OWASP Foundation, 2023*). From pilfering personal information to manipulating financial records, the attacker holds the power to wreak havoc on Costco's digital infrastructure. This intricate dance between probing tools and the website's vulnerabilities underscores the evolving nature of cyber threats. As organizations strive to fortify their digital defenses, understanding the nuances of such attacks becomes imperative. The scenario highlights the critical need for constant vigilance, robust security measures, and an unwavering commitment to staying one step ahead of those who seek to exploit the ever-expanding digital frontier.

SQL injection attacks, though relatively straightforward to implement, underscore the importance of robust development practices for prevention. Paradoxically, while preventing SQL injections is conceptually uncomplicated, real-world challenges often muddy the waters. Tight project deadlines, inexperienced developers, and the prevalence of legacy code contribute to variable code quality and security practices within a website's architecture. The critical issue lies

in the fact that even a single vulnerable field on a form or API endpoint if overlooked, can serve as a gateway for an attacker to compromise the entire system (Linode, 2023). As such, the security of a website is only as strong as its weakest link. To fortify against SQL injection attacks, organizations like Costco must prioritize secure coding practices, conduct regular security audits, and foster a culture of ongoing vigilance. In an era where cyber threats evolve rapidly, robust security measures are imperative to safeguard sensitive data and maintain the integrity of digital infrastructures.

Another attack pen testing technique we deployed was spear phishing. In a high-stakes scenario, Costco finds itself ensnared in a web of cyber threats as sophisticated spear phishing tactics are employed against the retail giant. The attackers, armed with advanced tools, embark on a strategic mission to infiltrate Costco's network by targeting specific individuals within the organization through deceptive emails.

Unlike traditional phishing campaigns that cast a wide net, spear phishing is a precise and targeted approach. In this narrative, the attackers craft highly convincing emails masquerading as entities familiar to the victims, leveraging their trust to coax sensitive data, fund transfers, or the unwitting download of malicious malware. The malevolent actors capitalize on the art of social engineering, exploiting inherent human tendencies such as the inclination to assist, respond to alarming cues, or trust superiors. In the complex landscape of corporate cybersecurity, we exploit the vulnerability of employees as a significant weak point (cisco, 2023). The unfortunate reality is that human error is identified as the primary cause of unplanned system downtime, emphasizes the critical need for continuous anti-phishing training within organizations. As Costco grapples with the aftermath of these targeted spear phishing attacks, the narrative underscores the importance of holistic security measures. Beyond technological defenses, cultivating a culture of

awareness and vigilance is paramount. The story serves as a stark reminder that, in the face of evolving cyber threats, organizations must be proactive in fortifying their defenses, conducting regular security audits, and providing ongoing training to ensure the resilience of their digital ecosystems.

The SQL injection and spear phishing attacks orchestrated against Costco's warehouse succeeded due to a combination of clever tactics, exploited vulnerabilities, and unsuspecting targets. In the case of SQL injection, the attackers identified and targeted a vulnerable Application Programming Interface (API) within Costco's network. Through automated tools, they systematically probed for weaknesses in the website's forms and API endpoints, injecting malicious SQL queries that the developers did not anticipate. This allowed the attackers to manipulate the database, gaining unauthorized access to sensitive information stored within.

Simultaneously, spear phishing tactics targeted specific individuals within Costco's workforce. The attackers, employing highly convincing emails impersonating trusted entities, leveraged social engineering tactics to manipulate employees into divulging sensitive information or unwittingly facilitating the download of malware. The success of spear phishing hinged on exploiting human vulnerabilities, such as the inclination to trust familiar figures or respond urgently to perceived authority.

The stolen data, a trove of sensitive corporate information, resided within Costco's database and various employee systems. The attackers, having successfully breached the initial layers of security, gained access to customer data, financial records, and potentially confidential business information. This breach not only resulted in immediate financial and data loss but also opened the door for more extensive and persistent intrusions. The significance of the stolen data

lies in the potential for financial fraud, identity theft, and the compromise of critical operational information.

To prevent future attacks of this nature, Costco should consider comprehensive security upgrades. This includes implementing rigorous data input validation practices to fortify against SQL injection attempts. Furthermore, continuous and targeted anti-phishing training for employees can enhance their resilience against social engineering tactics. Strengthening network defenses through advanced email security solutions and multi-factor authentication can add additional layers of protection. Regular security audits, timely patch management, and a culture of cybersecurity awareness are crucial components of an effective security posture that can mitigate the risks posed by evolving cyber threats.

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